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Exploring the potential of black liquor as a reinforcing agent for modern adobe brick composites

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INTRODUCTION

- Throughout history, earth has often been used as a construction material for urban and rural houses in Romania[1];
- In the Banat region, one of the most widespread ways of construction was using adobe bricks, molded into a rectangular shape and air-dried [2];
- Black liquor is an aqueous solution comprised of lignin and hemicellulose residues and the inorganic chemicals used in the kraft pulping process[3];
- The main objective of the present study was to explore the potential of black liquor as a reinforcing agent for adobe brick, aiming to obtain composites with improved mechanical properties.

Black liquor

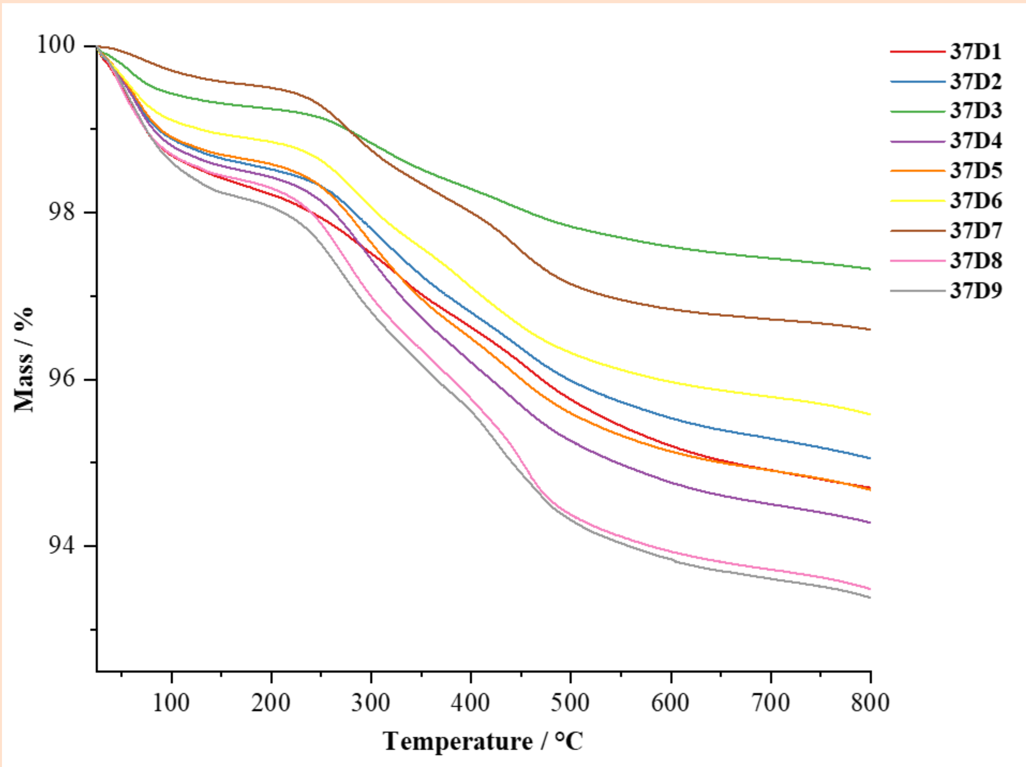
MATERIALS AND METHODS

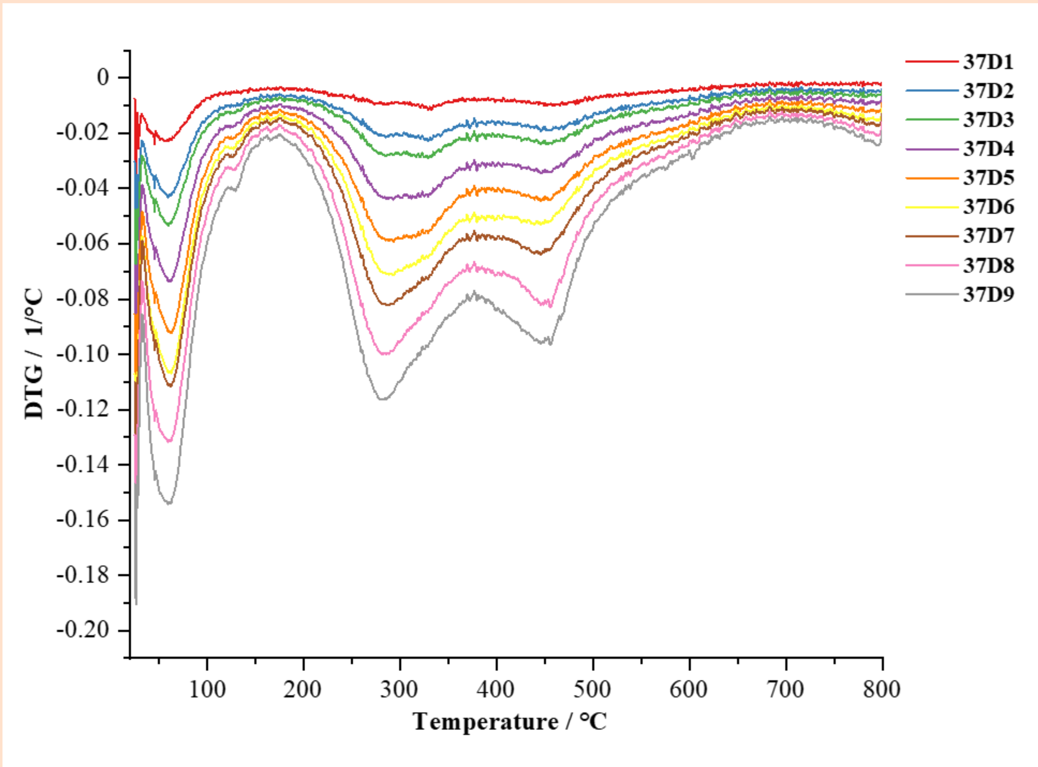
- Black liquor was purchased by *Eco Living Project S.R.L* from the Combinatul de Celuloză și Hârtie (CCH) Drobeta Turnu Severin, Romania, while the clay, shredded straw, and ecological plaster premix, marketed as ***Caminota***® were made available by Eco living project S.R.L.;
- Composite samples were prepared by combining ***Caminota***® premix with clay, shredded straw, and Black liquor to determine its potential as a reinforcing agent for adobe composites;
- The resulting composite samples were analyzed through thermogravimetric analysis, SEM, and FTIR analysis, while the influence of black liquor on their mechanical properties, namely compression and flexural strength, following EN 1015-11 [4] testing methodology.

RESULTS

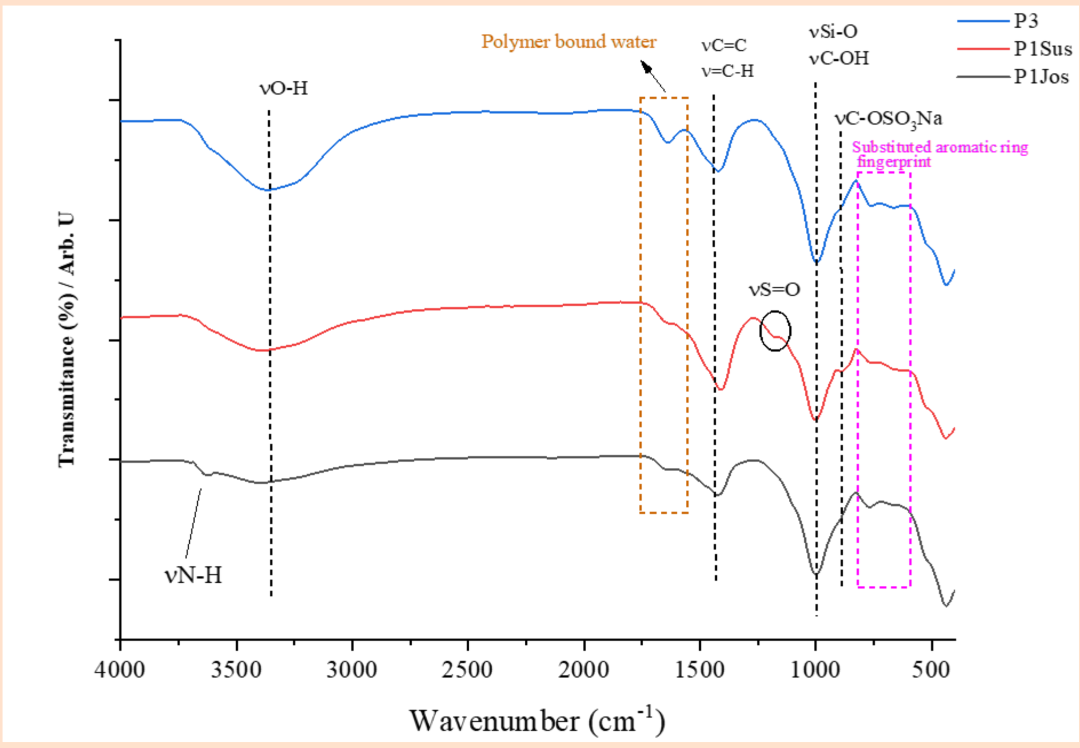
Adobe Composites

TG/DTG





FTIR



SEM

CONCLUSION

- The resulting samples show variations in physical appearance, both in color due to the presence of black liquor and in the integrity of the composite, some becoming brittle as they lose water;
- FTIR analysis indicates successful incorporation of Black liquor into the entire mass of the resulting composite;
- According to the thermal analysis, the first thermal processes of the DTG curve correspond to moisture losses, located in the temperature range 33-95 °C. These are followed by removal of the sulfonic group from the lignosulfonate, with an onset T of 230°C, and dehydroxylation of the clay minerals in the temperature range ≈380-520°C;
- The SEM analysis shows variations in the microscopic morphology of the samples. Furthermore, as the amount of black liquor increases, a better packing of the system can be observed; thus, the structure of the analyzed composite becomes much more uniform;
- To determine if black liquor can be used as a reinforcing agent, more extensive studies should be carried out regarding the impact of this compound on the compression and flexural strength of resulting composites, obtained samples should be tested at 1 week, 3, and 6 weeks after when casting the initial samples.

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